

Week 7 Weekly Report

Week 7 extended the selected models with counterfactual stress tests, VIX-focused interpretation, a key-free public-data updater, and a working Streamlit prototype. VIX remains a volatility and market-fear indicator; it is not relabeled as news sentiment.

1 Work Completed

- Measured VIX-related SHAP contribution for the selected Random Forest and Linear Regression models.
- Applied the required 50% volatility spike and absolute 2 percentage-point rate hike to the structural BSM model.
- Applied the same counterfactual inputs to both machine-learning pricing routes.
- Recorded the absence of a licensed or reproducible news-sentiment series instead of fabricating a feature.
- Built an online updater using JPM public chart data, Cboe VIX history, and FRED DGS1, with an explicit Week 2 fallback.
- Built and locally health-tested a Streamlit pricing and sensitivity prototype.

2 Deliverables

- Week7_Advanced_Analysis.ipynb
- sensitivity_analysis.py
- data_update.py
- Streamlit app.py
- tool_config.json
- stress, SHAP, and data-status outputs

3 Progress Summary

Week 7 is complete. The required structural stress scenarios and ML counterfactuals are reproducible. The online updater returned a successful snapshot during testing, and the Streamlit server health endpoint returned ok.

4 Experimental Results

Scenario	BSM	Approach 1	Approach 2
Base	99.2617	91.7072	0.0000
50% volatility spike	115.6683	92.0449	0.0000
2% rate hike	101.3317	94.5557	0.0000
Combined shock	116.9177	94.8010	0.0000

Figure 1 Structural chooser stress scenarios



5 Issues Encountered

- The first Yahoo Finance library request was rate-limited. The updater was changed to a public JPM chart endpoint plus Cboe and FRED sources, then retested successfully.
- A news-sentiment feature requires a defined source, timestamp policy, and leakage controls. These inputs were not present in Weeks 1 and 2, so sentiment remains a documented data gap.
- ML counterfactuals can combine inputs that are uncommon in the training sample; results must be read with the training-domain limits.

6 Plan for Week 8

- Package BSM and the best ML model into a self-contained final tool.
- Display an empirical proxy-error margin and model limitations.
- Prepare the deployment checklist, final report, presentation, and demo script.